

How the Subject-Specific Denominator Enters Equation 6

A note on the event contribution in Zhao et al.

The two delta symbols

Equation 6 weights each event contribution by the number of usable gaps for the subject who generated that event. The notation can be confusing because the equation uses both an uppercase and a lowercase delta:

- Δ_{ij} is the event indicator. It is the quantity divided by m_i^* .
- δ_i is the group indicator. It stays unchanged.

Thus, the substitution is $\Delta_{ij} \mapsto \Delta_{ij}/m_i^*$, not $\delta_i \mapsto \delta_i/m_i^*$. Equation 6 uses m_i^* , the paper's adjusted count of usable gaps, rather than the unadjusted m_i .

Applying the substitution to the second term

Write the event-level part of the second term in Equation 5 as

$$q_{i'j'} = \Delta_{i'j'} L(\tilde{t}_{i'j'}) \left\{ \delta_{i'} - \frac{Y_1(\tilde{t}_{i'j'})}{Y(\tilde{t}_{i'j'})} \right\}.$$

The second term is then

$$\sum_{i'=1}^n \sum_{j'=1}^{m_{i'}} q_{i'j'} \frac{Y_{g_i}(\tilde{t}_{i'j'})}{Y(\tilde{t}_{i'j'})}.$$

In Equation 6, the event-level quantity is replaced by its Wang–Chang weighted version and divided by the gap count of its source subject:

$$\sum_{i'=1}^n \sum_{j'=1}^{m_{i'}} \frac{q_{i'j'}^*}{m_{i'}^*} \frac{Y_{g_i}^*(\tilde{t}_{i'j'})}{Y^*(\tilde{t}_{i'j'})},$$

where

$$q_{i'j'}^* = \Delta_{i'j'} L^*(\tilde{t}_{i'j'}) \left\{ \delta_{i'} - \frac{Y_1^*(\tilde{t}_{i'j'})}{Y^*(\tilde{t}_{i'j'})} \right\}.$$

For any fixed subject i' , the value $m_{i'}^*$ is constant over that subject's event index j' . So that subject's contribution can be written as

$$\frac{1}{m_{i'}^*} \sum_{j'=1}^{m_{i'}} q_{i'j'}^* \frac{Y_{g_i i}^*(\tilde{t}_{i'j'})}{Y^*(\tilde{t}_{i'j'})}.$$

This is the key step: $1/m_{i'}^*$ is a denominator for each source subject's inner sum. Since different subjects can have different values of $m_{i'}^*$, there is no single denominator to pull in front of the full double sum.

Why the first term looks different

The first term is indexed over events from one focal subject i . Its factor $1/m_i^*$ is therefore common to that entire sum over j :

$$\frac{1}{m_i^*} \sum_{j=1}^{m_i} (\text{weighted event contribution for subject } i).$$

In the second term, the outer sum runs over many source subjects i' , each with its own denominator $m_{i'}^*$. You can factor the denominator outside each subject's inner sum, but not outside the sum over subjects.

The package represents these subject-specific weights inside the weighted risk and event increments, so the factor $1/m_i^*$ may not appear as a separate denominator in every later expression.